

Rademacher Transseries for the Partition Numbers and Their Inverse

**All-order arithmetic sectors, coefficient formulae,
Lambert–Lagrange reversion, and phase-locked inversion**

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Abstract

Let $p(n)$ be the unrestricted partition number. The exact Rademacher series is reorganized here as a complete arithmetic transseries. In the natural shifted variable

$$x = n - \frac{1}{24}, \quad \mu = \pi\sqrt{\frac{2x}{3}},$$

each fixed Rademacher sector is exactly an exponential multiplied by a *two-term* rational prefactor. Re-expansion in the conventional variable $n^{-1/2}$ produces, for every sector (k, σ) with $k \geq 1$ and $\sigma \in \{+1, -1\}$, a convergent all-order coefficient series. We give three equivalent general formulae for its coefficients: an analytic generating function, a complete-Bell-polynomial expression, and the finite sum

$$C_{k, \sigma; r} = \frac{1}{(-4\sqrt{6})^r} \sum_{j=0}^{\lfloor (r+1)/2 \rfloor} \binom{r+1}{j} \frac{r+1-j}{(r+1-2j)!} \left(\frac{\sigma\pi}{6k}\right)^{r-2j}.$$

The case $(k, \sigma) = (1, +1)$ is the known O’Sullivan coefficient formula; the displayed expression supplies its sector-wide extension.

Because p is a sequence, “the inverse” first requires a convention. We construct a canonical real-analytic Rademacher interpolation $\mathcal{P}(\nu)$ that agrees with $p(n)$ at every integer and is strictly increasing for all sufficiently large ν . Its inverse $\mathcal{N}(y)$ therefore determines the generalized inverse of the sequence by $p_{\mathbb{Z}}^{\langle -1 \rangle}(y) = \lceil \mathcal{N}(y) \rceil$ for large y . The dominant inverse is anchored by the exact Lambert carrier

$$\rho(y) = -2W_{-1}\left(-\frac{1}{2}\sqrt{\frac{\kappa}{y}}\right), \quad \kappa = \frac{\pi^2}{6\sqrt{3}},$$

and the full algebraic correction is generated by Lagrange–Bürmann and ordinary Bell polynomials. Its coefficients are given both by finite coefficient extraction and by a triangular recurrence. The complete nonperturbative inverse is then obtained by a second Lagrange inversion. A finite multi-index formula gives every product-sector coefficient, including all derivatives of the periodic Dedekind-sum amplitudes.

Two structural points are essential. First, the positive Rademacher actions $1 - 1/k$ accumulate at 1, so the completed object is an arithmetically indexed, paired-summation transseries rather than a finite-rank Hahn series. Second, the Lambert carrier and the dominant-core inverse differ by $3/\pi^2 + o(1)$ in the original index. This constant must be resummed before the periodic amplitudes are evaluated; otherwise their phases are wrong at leading nonperturbative order. The leading inverse correction is the parity chirp

$$\mathcal{N}(y) - N_0(y) \sim -\frac{3\sqrt{\kappa}}{\pi^2\sqrt{2}} \frac{\cos(\pi N_0(y))}{\sqrt{y}}.$$

Explicit remainder transfer principles, implementation algorithms, and high-precision numerical checks are included.

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1 Introduction

The unrestricted partition function $p(n)$ counts non-increasing finite sequences of positive integers with sum n ; it is OEIS A000041 [13]. Its generating function is Euler's product

$$\sum_{n \geq 0} p(n)q^n = \prod_{m \geq 1} \frac{1}{1 - q^m}, \quad |q| < 1. \quad (1.1)$$

Hardy and Ramanujan proved the celebrated first-order estimate [1]

$$p(n) \sim \frac{1}{4\sqrt{3}n} \exp\left(\pi\sqrt{\frac{2n}{3}}\right). \quad (1.2)$$

Rademacher's refinement is not merely an asymptotic series: it is a convergent sum indexed by positive integers k , and each k carries a finite exponential sum built from Dedekind sums [2, 3, 4, 5, 8]. Consequently the genuinely complete asymptotic object has three interacting structures:

- (i) an algebraic half-power expansion in $n^{-1/2}$;
- (ii) exponentially small Rademacher sectors with actions $1 \mp 1/k$;
- (iii) periodic arithmetic amplitudes, which become chirped functions of $\log y$ after inversion.

Keeping only item (i) gives the usual Poincaré expansion. It suppresses all arithmetic oscillation and therefore cannot describe the exponentially accurate inverse.

This article develops the direct and inverse expansions in a form compatible with the coefficient calculus based on Bell polynomials, Faà di Bruno composition, and Lagrange–Bürmann inversion [11, 12]. Formal identities are kept separate from analytic error statements. In particular, an exact convergent Rademacher expression supplies the nonperturbative completion, while finite coefficient extraction gives every formal coefficient.

1.1 What is meant by “full”

For fixed k , the two exponentials $e^{\pm\mu/k}$ are perfectly ordinary transseries sectors. Across all k , however, the actions

$$1 - \frac{1}{2}, 1 - \frac{1}{3}, 1 - \frac{1}{4}, \dots$$

approach 1 from below, while the actions $1 + 1/k$ approach 1 from above. The latter set has no least element after restricting to an infinite tail. In addition, the two signs for a given large k must be paired before summing: the large- k convergence is produced by cancellation between them. Thus the complete object is not a classical grid-based transseries with globally well-ordered support.

We use the following precise replacement.

Definition 1.1 (Rademacher transseries). A Rademacher transseries is a compatible family of finite- K resolved sector expansions, together with the paired convergent Rademacher tail. Every action cutoff strictly below 1 contains only finitely many resolved primary sectors. At and beyond the accumulation action 1, the paired tail is retained as an analytic coefficient, or is resolved transasymptotically by increasing K .

This definition preserves all information in the exact series and avoids an illegitimate rearrangement of a conditionally structured infinite sector sum.

1.2 Main results at a glance

The direct expansion has an exceptionally simple shifted form. Put

$$c := \pi\sqrt{\frac{2}{3}} = \frac{\pi\sqrt{6}}{3}, \quad x := n - \frac{1}{24}, \quad \mu := c\sqrt{x}, \quad \kappa := \frac{c^2}{4\sqrt{3}} = \frac{\pi^2}{6\sqrt{3}}. \quad (1.3)$$

For every $k \geq 1$ and $\sigma = \pm 1$, the (k, σ) sector is

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}x} e^{\sigma\mu/k} \left(1 - \frac{\sigma k}{\mu}\right). \quad (1.4)$$

This is exact. Re-expanding it about unshifted $n = \infty$ gives the all-order coefficients stated in the abstract.

For inversion, the dominant positive sector is

$$D(\mu) = \kappa \frac{e^\mu}{\mu^2} \left(1 - \frac{1}{\mu}\right). \quad (1.5)$$

The Lambert carrier solves the simpler equation $y = \kappa e^\rho / \rho^2$. The exact dominant-core inverse μ_0 solves $y = D(\mu_0)$ and has a convergent expansion

$$\mu_0 = \rho + \frac{1}{\rho} + \frac{5}{2\rho^2} + \frac{13}{3\rho^3} + \frac{53}{12\rho^4} - \frac{14}{5\rho^5} + \cdots. \quad (1.6)$$

All remaining sectors are reverted by a universal second Lagrange formula.

The article gives explicit general coefficient formulae at every level:

1. $C_{k,\sigma;r}$ for every direct Rademacher sector, in finite-sum and Bell-polynomial form;
2. $P_m(\log L)$ for the elementary logarithmic expansion of the Lambert carrier, in signed-Stirling-number form;
3. a_m for the dominant-core correction, by finite Lagrange extraction and a triangular ordinary-Bell recurrence;
4. V_m for every product of nonperturbative inverse sectors, by a finite multi-index Lagrange formula.

2 Arithmetic and coefficient conventions

2.1 Dedekind sums and Rademacher amplitudes

For real u , let the sawtooth function be

$$((u)) := \begin{cases} u - \lfloor u \rfloor - \frac{1}{2}, & u \notin \mathbb{Z}, \\ 0, & u \in \mathbb{Z}. \end{cases} \quad (2.1)$$

For $k \geq 1$ and $h \in \mathbb{Z}$, the Dedekind sum is

$$s(h, k) := \sum_{r=1}^{k-1} \left(\left(\frac{r}{k} \right) \right) \left(\left(\frac{hr}{k} \right) \right). \quad (2.2)$$

The Rademacher amplitude at an integer n may be written either as a complex exponential sum or, because conjugate terms pair, as the real cosine sum

$$A_k(n) = \sum_{\substack{0 \leq h < k \\ (h,k)=1}} \exp\left(\pi i s(h, k) - \frac{2\pi i n h}{k}\right), \quad (2.3)$$

$$= \sum_{\substack{1 \leq h \leq k \\ (h,k)=1}} \cos\left(\pi s(h, k) - \frac{2\pi n h}{k}\right). \quad (2.4)$$

The convention gives $A_1(n) = 1$. The elementary bound

$$|A_k(n)| \leq \varphi(k) \leq k \quad (2.5)$$

is sufficient for the convergence estimates used below.

The first amplitudes are

$$\begin{aligned} A_1(n) &= 1, \\ A_2(n) &= (-1)^n, \\ A_3(n) &= 2 \cos\left(\frac{\pi}{18} - \frac{2\pi n}{3}\right), \\ A_4(n) &= 2 \cos\left(\frac{\pi}{8} - \frac{\pi n}{2}\right). \end{aligned} \tag{2.6}$$

No asymptotic averaging of these amplitudes is performed: their arithmetic phase is part of the answer.

2.2 Bell polynomials

The partial exponential Bell polynomials $B_{n,j}$ are defined by

$$B_{n,j}(x_1, x_2, \dots) = n! \sum_{\substack{m_1+m_2+\dots=j \\ m_1+2m_2+\dots=n}} \prod_{r \geq 1} \frac{1}{m_r!} \left(\frac{x_r}{r!}\right)^{m_r}. \tag{2.7}$$

The complete polynomial is $B_n = \sum_{j=0}^n B_{n,j}$. The basic exponential identity is

$$\exp\left(\sum_{r \geq 1} x_r \frac{t^r}{r!}\right) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) \frac{t^n}{n!}. \tag{2.8}$$

For an ordinary series $U(t) = \sum_{r \geq 1} u_r t^r$, write

$$\widehat{B}_{n,j}(u_1, u_2, \dots) := [t^n] U(t)^j. \tag{2.9}$$

Thus $\widehat{B}_{0,0} = 1$, $\widehat{B}_{n,0} = 0$ for $n > 0$, and

$$\widehat{B}_{n,j}(u) = \frac{j!}{n!} B_{n,j}(1!u_1, 2!u_2, \dots). \tag{2.10}$$

2.3 Signed Stirling numbers

We use signed Stirling numbers of the first kind $s(m, j)$:

$$z(z-1) \cdots (z-m+1) = \sum_{j=0}^m s(m, j) z^j. \tag{2.11}$$

Equivalently,

$$\log(1+z)^j = j! \sum_{m \geq j} s(m, j) \frac{z^m}{m!}. \tag{2.12}$$

2.4 Coefficient extraction and Lagrange–Bürmann

If $g(z) = z\phi(g(z))$ with $\phi(0)$ invertible, then

$$[z^m]g(z) = \frac{1}{m} [w^{m-1}] \phi(w)^m, \quad m \geq 1. \tag{2.13}$$

We repeatedly introduce a bookkeeping parameter z , apply (2.13), and finally set $z = 1$. In each application the resulting coefficient at a prescribed algebraic or exponential order is a finite sum.

3 The exact Rademacher decomposition

3.1 Rademacher's convergent series

Theorem 3.1 (Rademacher [2, 3, 14]). *For every integer $n \geq 1$,*

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n) \sqrt{k} \frac{d}{dn} \left[\frac{\sinh\left(\frac{c}{k} \sqrt{n - \frac{1}{24}}\right)}{\sqrt{n - \frac{1}{24}}} \right], \quad (3.1)$$

where $c = \pi\sqrt{2/3}$ and the series converges when summed in increasing k .

The square root of k in (3.1) is important. Some modern normalizations absorb it into the definition of the arithmetic sum; throughout this article A_k means (2.3)–(2.4).

3.2 Exact two-exponential sectors

Lemma 3.2 (Elementary derivative). *For $a > 0$ and $x > 0$,*

$$\frac{d}{dx} \left(x^{-1/2} \sinh(a\sqrt{x}) \right) = \frac{1}{4x^{3/2}} \left[(a\sqrt{x} - 1)e^{a\sqrt{x}} + (a\sqrt{x} + 1)e^{-a\sqrt{x}} \right]. \quad (3.2)$$

Proof. Differentiate the product $x^{-1/2} \sinh(a\sqrt{x})$, replace hyperbolic functions by exponentials, and collect the two signs. No asymptotic expansion is involved. $\square$

Theorem 3.3 (Exact sector formula). *Let $x = n - 1/24$ and $\mu = c\sqrt{x}$. Define, for $\sigma = \pm 1$,*

$$P_{k,\sigma}(n) := \frac{A_k(n)}{4\sqrt{3k}x} e^{\sigma\mu/k} \left(1 - \frac{\sigma k}{\mu} \right). \quad (3.3)$$

Then Rademacher's series is exactly

$$p(n) = \sum_{k \geq 1} (P_{k,+}(n) + P_{k,-}(n)), \quad (3.4)$$

where the two signs are paired before the sum over k is taken. Equivalently, with

$$g(z) := \cosh z - \frac{\sinh z}{z}, \quad g(0) := 0, \quad (3.5)$$

we have the paired form

$$P_{k,+}(n) + P_{k,-}(n) = \frac{A_k(n)}{2\sqrt{3k}x} g\left(\frac{\mu}{k}\right). \quad (3.6)$$

Proof. Insert (3.2) with $a = c/k$ into (3.1). Since

$$\frac{\sqrt{k}}{\pi\sqrt{2}} \frac{1}{4x^{3/2}} \frac{\mu}{k} = \frac{1}{4\sqrt{3k}x},$$

the two exponentials give (3.3). Their sum is (3.6). $\square$

Remark 3.4 (Why pairing is mandatory). For fixed k , splitting into $P_{k,+}$ and $P_{k,-}$ is harmless. For $k \rightarrow \infty$, each split term contains a component proportional to k/μ ; the cancellation of those components occurs only in their sum. The paired kernel satisfies $g(\mu/k) = \mu^2/(3k^2) + O(k^{-4})$, which restores convergence. Thus (3.4) specifies an order of summation, not an absolutely rearrangeable double series.

3.3 Dominant sector

The positive $k = 1$ sector is

$$P_{1,+}(n) = \frac{e^\mu}{4\sqrt{3}x} \left(1 - \frac{1}{\mu}\right) = \kappa \frac{e^\mu}{\mu^2} \left(1 - \frac{1}{\mu}\right). \quad (3.7)$$

The negative $k = 1$ sector is smaller by $e^{-2\mu}$ relative to it. The positive $k = 2$ sector is smaller by $e^{-\mu/2}$ and is the first arithmetic correction. This immediately explains why the shifted approximation (3.7) is exponentially more accurate than the elementary Hardy–Ramanujan formula (1.2).

4 The complete direct Rademacher transseries

4.1 Resolved actions and exact relative factorization

Let

$$N(\mu) := \frac{1}{24} + \frac{3\mu^2}{2\pi^2}, \quad (4.1)$$

so that $\mu(N(\mu)) = \mu$. The same notation will later be used for the real interpolation. Relative to (3.7), define

$$\alpha_{k,\sigma} := 1 - \frac{\sigma}{k}, \quad (4.2)$$

$$r_{k,\sigma}(\mu) := \frac{A_k(N(\mu))}{\sqrt{k}} \frac{1 - \sigma k/\mu}{1 - 1/\mu}. \quad (4.3)$$

At integer arguments, $A_k(N(\mu))$ is interpreted on the corresponding lattice point. Then, for any finite K ,

$$\frac{p(n)}{P_{1,+}(n)} = 1 + r_{1,-}(\mu)e^{-2\mu} + \sum_{k=2}^K \sum_{\sigma=\pm 1} r_{k,\sigma}(\mu)e^{-\alpha_{k,\sigma}\mu} + \mathcal{R}_{>K}(\mu). \quad (4.4)$$

The actions are

$$\alpha_{k,+} = 1 - \frac{1}{k}, \quad \alpha_{k,-} = 1 + \frac{1}{k}. \quad (4.5)$$

4.2 The paired tail and action accumulation

The exact non-dominant relative correction can be written without any split large- k sectors:

$$\mathcal{R}(\mu) = \frac{1 + 1/\mu}{1 - 1/\mu} e^{-2\mu} + \frac{2e^{-\mu}}{1 - 1/\mu} \sum_{k=2}^{\infty} \frac{A_k(N(\mu))}{\sqrt{k}} g\left(\frac{\mu}{k}\right). \quad (4.6)$$

Thus

$$p(n) = \kappa \frac{e^\mu}{\mu^2} \left(1 - \frac{1}{\mu}\right) (1 + \mathcal{R}(\mu)). \quad (4.7)$$

Equation (4.6) is the analytic completion of the resolved transseries.

Lemma 4.1 (A useful kernel bound). *For $z \geq 0$,*

$$0 \leq g(z) \leq \frac{z^2}{3} \cosh z \leq \frac{z^2}{3} e^z. \quad (4.8)$$

Proof. The Taylor series is

$$g(z) = \sum_{m \geq 1} \frac{2m}{(2m+1)!} z^{2m}.$$

Since $2m/(2m+1)! \leq 1/(3(2m-2)!)$ for $m \geq 1$, termwise comparison gives the first upper bound. The second is immediate. $\square$

Theorem 4.2 (Explicit finite- K tail bound). *Let $\mu > 1$ and $K \geq 1$. For the real cosine amplitudes, and hence at every integer n , the paired tail in (4.4) satisfies*

$$|\mathcal{R}_{>K}(\mu)| \leq \frac{4\mu^2}{3(1-1/\mu)\sqrt{K}} \exp\left[-\left(1 - \frac{1}{K+1}\right)\mu\right]. \quad (4.9)$$

Proof. By (2.5), (4.8), and $\sum_{k>K} k^{-3/2} \leq 2/\sqrt{K}$,

$$\begin{aligned} |\mathcal{R}_{>K}| &\leq \frac{2e^{-\mu}}{1-1/\mu} \sum_{k>K} \sqrt{k} g(\mu/k) \\ &\leq \frac{2\mu^2 e^{-\mu}}{3(1-1/\mu)} \sum_{k>K} k^{-3/2} e^{\mu/k} \\ &\leq \frac{4\mu^2}{3(1-1/\mu)\sqrt{K}} e^{-\mu+\mu/(K+1)}. \end{aligned}$$

□

The actions $1 - 1/k$ form an increasing sequence with limit 1, but the actions $1 + 1/k$ decrease to 1. Hence the complete primary action set is not globally well ordered. Formula (4.9) is the correct replacement for a formal assertion that an infinite list of split sectors may be freely sorted. For every cutoff $\eta < 1$, only finitely many positive sectors satisfy $\alpha_{k,+} \leq \eta$; at the accumulation action, one uses the paired tail.

5 All-order coefficients in the unshifted variable

The shifted sector formula (3.3) is exact and elementary. It is often convenient, however, to normalize by the conventional Hardy–Ramanujan exponential $e^{\sigma c \sqrt{n}/k}$ and expand in $n^{-1/2}$. This section gives all coefficients in three equivalent forms.

5.1 Analytic generating function

Set

$$t := n^{-1/2}, \quad \delta := \frac{1}{24}, \quad \beta_{k,\sigma} := \frac{\sigma c}{k}. \quad (5.1)$$

Theorem 5.1 (Sector generating function). *For every fixed (k, σ) ,*

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}n} e^{\sigma c \sqrt{n}/k} \sum_{r \geq 0} C_{k,\sigma;r} n^{-r/2}, \quad (5.2)$$

where

$$\sum_{r \geq 0} C_{k,\sigma;r} t^r = \frac{\exp\left[\frac{\sigma c}{kt} \left(\sqrt{1 - \frac{t^2}{24}} - 1\right)\right]}{1 - \frac{t^2}{24}} \left(1 - \frac{\sigma kt}{c\sqrt{1 - \frac{t^2}{24}}}\right). \quad (5.3)$$

The right-hand side is analytic at $t = 0$ and has radius of analyticity at least $\sqrt{24}$. Consequently (5.2) is convergent for every integer $n \geq 1$ when k and σ are fixed.

Proof. Write $x = n(1 - t^2/24)$ and $\mu = (c/t)\sqrt{1 - t^2/24}$ in (3.3). Factor out $e^{\sigma c/(kt)}$ and $1/n$. The apparent singularity in the remaining exponent is removable because $\sqrt{1 - t^2/24} - 1 = O(t^2)$. The nearest branch points are at $t = \pm\sqrt{24}$. □

5.2 Finite closed form

Define

$$b_{k,\sigma} := \frac{\sigma\pi}{6k}. \quad (5.4)$$

Theorem 5.2 (Finite coefficient formula). *For every $r \geq 0$,*

$$C_{k,\sigma;r} = \frac{1}{(-4\sqrt{6})^r} \sum_{j=0}^{\lfloor (r+1)/2 \rfloor} \binom{r+1}{j} \frac{r+1-j}{(r+1-2j)!} b_{k,\sigma}^{r-2j}. \quad (5.5)$$

Proof. A complete residue derivation is given in [Section A](#). The key substitution is $u = -t/(4\sqrt{6})$ followed by the Catalan change of variable $T = u(1 + T^2)$. Lagrange residue transfer reduces the coefficient to

$$\frac{r+1}{b_{k,\sigma}} [T^{r+1}] e^{b_{k,\sigma} T} (1 + T^2)^r,$$

whose binomial expansion is exactly (5.5). □

The first coefficients, with $b = b_{k,\sigma}$, are

$$\begin{aligned} C_0 &= 1, \\ C_1 &= -\frac{\sqrt{6}(b^2 + 2)}{24b}, \\ C_2 &= \frac{b^2 + 12}{192}, \\ C_3 &= -\frac{\sqrt{6}(b^4 + 36b^2 + 72)}{13824b}, \\ C_4 &= \frac{b^4 + 80b^2 + 720}{221184}, \\ C_5 &= -\frac{\sqrt{6}(b^6 + 150b^4 + 3600b^2 + 7200)}{26542080b}. \end{aligned} \quad (5.6)$$

5.3 Complete-Bell-polynomial form

Let

$$h_{2m-1}^{(k,\sigma)} := \frac{\sigma c}{k} \binom{1/2}{m} \left(-\frac{1}{24}\right)^m, \quad h_{2m}^{(k,\sigma)} := 0. \quad (5.7)$$

Define

$$D_r^{(\lambda;k,\sigma)} := \sum_{q=0}^{\lfloor r/2 \rfloor} \frac{(\lambda)_q}{q! 24^q} \frac{B_{r-2q}(1! h_1^{(k,\sigma)}, \dots, (r-2q)! h_{r-2q}^{(k,\sigma)})}{(r-2q)!}, \quad (5.8)$$

with $D_r^{(\lambda)} = 0$ for $r < 0$.

Proposition 5.3 (Bell-polynomial coefficient calculus). *The sector coefficients are*

$$C_{k,\sigma;r} = D_r^{(1;k,\sigma)} - \frac{\sigma k}{c} D_{r-1}^{(3/2;k,\sigma)}. \quad (5.9)$$

Proof. The exponent in (5.3) is $\sum_{j \geq 1} h_j t^j$. Apply (2.8), multiply first by $(1 - t^2/24)^{-1}$ and then by $1 - (\sigma k/c)t(1 - t^2/24)^{-1/2}$. Cauchy convolution gives (5.8)–(5.9). □

Formula (5.9) is often preferable in symbolic systems because it uses only standard composition primitives. Formula (5.5) is usually faster for numerical coefficient generation.

5.4 Recovery of the ordinary Poincare expansion

For the dominant positive sector, $b_{1,+} = \pi/6$. Set

$$\omega_r := C_{1,+;r}. \quad (5.10)$$

Then (5.5) becomes

$$\omega_r = \frac{1}{(-4\sqrt{6})^r} \sum_{j=0}^{\lfloor (r+1)/2 \rfloor} \binom{r+1}{j} \frac{r+1-j}{(r+1-2j)!} \left(\frac{\pi}{6}\right)^{r-2j}, \quad (5.11)$$

which is the O'Sullivan formula [6, 7]. Since every other fixed- k sector is exponentially small relative to $P_{1,+}$, one obtains, for each fixed R ,

$$p(n) = \frac{e^{c\sqrt{n}}}{4\sqrt{3}n} \left(\sum_{r=0}^{R-1} \omega_r n^{-r/2} + O_R(n^{-R/2}) \right). \quad (5.12)$$

Remark 5.4 (Convergent sector, asymptotic total). The infinite series $\sum_r \omega_r n^{-r/2}$ converges to the single sector $P_{1,+}(n)$ for $n \geq 1$; it does not converge to $p(n)$. The difference $p - P_{1,+}$ is beyond all algebraic orders and is supplied by the remaining Rademacher sectors. Thus (5.12) is an asymptotic statement about the complete partition function, even though its algebraic coefficient series has a convergent sectorial sum.

The first dominant coefficients are

$$\begin{aligned} \omega_0 &= 1, \\ \omega_1 &= -\frac{\sqrt{6}(\pi^2 + 72)}{144\pi}, \\ \omega_2 &= \frac{\pi^2 + 432}{6912}, \\ \omega_3 &= -\frac{\sqrt{6}(\pi^4 + 1296\pi^2 + 93312)}{2985984\pi}, \\ \omega_4 &= \frac{\pi^4 + 2880\pi^2 + 933120}{286654464}. \end{aligned} \quad (5.13)$$

6 A canonical inverse for a discrete sequence

6.1 Real-analytic Rademacher interpolation

A sequence has no unique continuous inverse. To retain the exponentially small arithmetic information, an interpolation must also retain the Rademacher amplitudes. The cosine representation provides a natural choice.

Definition 6.1 (Real Rademacher amplitude). For real or complex ν , define

$$\mathcal{A}_k(\nu) := \sum_{\substack{1 \leq h \leq k \\ (h,k)=1}} \cos\left(\pi s(h, k) - \frac{2\pi h\nu}{k}\right). \quad (6.1)$$

Then $\mathcal{A}_k(n) = A_k(n)$ for every integer n .

The first off-lattice amplitudes are

$$\mathcal{A}_2(\nu) = \cos(\pi\nu), \quad \mathcal{A}_3(\nu) = 2 \cos(\pi\nu) \cos\left(\frac{\pi}{18} + \frac{\pi\nu}{3}\right). \quad (6.2)$$

At integer $\nu = n$, the second expression reduces to the simpler lattice form in (2.6); that lattice-only simplification must not be used when an inverse phase is evaluated at the generally nonintegral point $N_0(y)$.

For real $\nu > 1/24$, set

$$\mathcal{P}(\nu) := \frac{1}{\pi\sqrt{2}} \sum_{k \geq 1} \mathcal{A}_k(\nu) \sqrt{k} \frac{d}{d\nu} \left[\frac{\sinh\left(\frac{c}{k} \sqrt{\nu - \frac{1}{24}}\right)}{\sqrt{\nu - \frac{1}{24}}} \right]. \quad (6.3)$$

Proposition 6.2 (Interpolation and eventual monotonicity). *The series (6.3) converges locally uniformly on $\nu > 1/24$ after the two signs are paired for each k . It defines a real-analytic function satisfying $\mathcal{P}(n) = p(n)$ for every integer $n \geq 1$. Moreover, there exists ν_0 such that $\mathcal{P}'(\nu) > 0$ for $\nu \geq \nu_0$.*

Proof. The paired k th term is (3.6) with A_k replaced by $\mathcal{A}_k$. On compact subsets, $|\mathcal{A}_k| \leq k$ and $g(\mu/k) = O(k^{-2})$, so the terms are $O(k^{-3/2})$ uniformly. The same argument applies after differentiation; derivatives of $\mathcal{A}_k$ have the same $O(k)$ trivial bound. This proves local uniform convergence and permits termwise differentiation. Equality at integers follows from Theorem 3.1.

The dominant term is $D(\mu)$ from (3.7). Its logarithmic derivative with respect to μ is

$$1 - \frac{2}{\mu} + \frac{1}{\mu(\mu-1)} = 1 + O(\mu^{-1}), \quad (6.4)$$

and $d\mu/d\nu = c^2/(2\mu) > 0$. The differentiated non-dominant part is, by the same estimates as Theorem 4.2, a polynomial in μ times $e^{-\mu/2}$ relative to the dominant derivative. Hence $\mathcal{P}'(\nu) = D'(\nu)(1+o(1)) > 0$. $\square$

6.2 Smooth and discrete inverses

Let $\mathcal{N}(y)$ denote the inverse branch of $\mathcal{P}$ on $[\nu_0, \infty)$. For the sequence, define the generalized inverse

$$p_{\mathbb{Z}}^{\langle -1 \rangle}(y) := \min\{n \in \mathbb{N} : p(n) \geq y\}. \quad (6.5)$$

Corollary 6.3 (Exact rounding relation). *For all sufficiently large real y ,*

$$\boxed{p_{\mathbb{Z}}^{\langle -1 \rangle}(y) = \lceil \mathcal{N}(y) \rceil}. \quad (6.6)$$

In particular, if $y = p(n)$ and n is large, then $\mathcal{N}(y) = n$.

Proof. On the interval of monotonicity, $\mathcal{P}$ is increasing and agrees with p at integers. If $p(n-1) < y \leq p(n)$, then $n-1 < \mathcal{N}(y) \leq n$. $\square$

Remark 6.4 (Dependence on interpolation). Other smooth interpolations can agree with $p(n)$ on the lattice and differ between integers. The algebraic inverse expansion is interpolation-independent; nonperturbative phases are not. Definition 6.1 is canonical for this article because it is obtained by extending the exact cosine amplitudes already present in Rademacher's formula. The integer generalized inverse is, of course, independent of that choice after rounding.

7 Lambert carrier and complete algebraic inversion

7.1 Dominant-core equation

For the interpolation, put

$$N(\mu) = \frac{1}{24} + \frac{3\mu^2}{2\pi^2} \quad (7.1)$$

and replace A_k by $\mathcal{A}_k$ in (4.6). The exact inverse equation is

$$y = \kappa \frac{e^\mu}{\mu^2} \left(1 - \frac{1}{\mu}\right) (1 + \mathcal{R}(\mu)). \quad (7.2)$$

Write

$$L := \log \frac{y}{\kappa}. \quad (7.3)$$

The dominant-core phase function is

$$F_0(\mu) := \mu - 2 \log \mu + \log \left(1 - \frac{1}{\mu}\right). \quad (7.4)$$

It satisfies

$$F'_0(\mu) = 1 - \frac{2}{\mu} + \frac{1}{\mu(\mu-1)}, \quad F'_0(\mu) = 1 + O(\mu^{-1}). \quad (7.5)$$

Definition 7.1 (Lambert carrier and dominant core). The Lambert carrier $\rho = \rho(y)$ is the large solution of

$$\rho - 2 \log \rho = L. \quad (7.6)$$

The dominant-core variable $\mu_0 = \mu_0(y)$ is the large solution of

$$F_0(\mu_0) = L. \quad (7.7)$$

Finally,

$$N_0(y) := \frac{1}{24} + \frac{3\mu_0(y)^2}{2\pi^2}. \quad (7.8)$$

7.2 Exact Lambert representation

Proposition 7.2 (Lambert W_{-1} carrier). *For sufficiently large y ,*

$$\rho(y) = -2W_{-1}\left(-\frac{1}{2}\sqrt{\frac{\kappa}{y}}\right). \quad (7.9)$$

The branch W_{-1} is forced by $\rho \rightarrow +\infty$ as $y \rightarrow +\infty$ [9, 10].

Proof. Equation (7.6) is equivalent to $\rho^2 e^{-\rho} = \kappa/y$. Setting $w = -\rho/2$ gives $w e^w = -\frac{1}{2}\sqrt{\kappa/y}$. The principal branch tends to zero; W_{-1} tends to $-\infty$. $\square$

7.3 Elementary logarithmic expansion and signed-Stirling formula

Let $\ell := \log L$. The carrier has the formal expansion

$$\rho = L + 2\ell + \sum_{m \geq 1} \frac{P_m(\ell)}{L^m}. \quad (7.10)$$

Theorem 7.3 (Closed formula for the Lambert polynomials). *For $m \geq 1$,*

$$P_m(\ell) = \frac{1}{m!} \sum_{j=1}^m 2^j (j-1)! s(m, j) \binom{m}{j-1} (2\ell)^{m-j+1}. \quad (7.11)$$

The first four are

$$\begin{aligned} P_1(\ell) &= 4\ell, \\ P_2(\ell) &= -4\ell(\ell-2), \\ P_3(\ell) &= \frac{8\ell}{3}(2\ell^2 - 9\ell + 6), \\ P_4(\ell) &= -\frac{8\ell}{3}(3\ell^3 - 22\ell^2 + 36\ell - 12). \end{aligned} \quad (7.12)$$

Proof. Put $v = \rho - L - 2\ell$ and $q = L^{-1}$. Equation (7.6) becomes

$$v = 2 \log(1 + q(2\ell + v)). \quad (7.13)$$

Introduce $v = z 2 \log(1 + q(2\ell + v))$. Lagrange–Bürmann gives

$$v = \sum_{j \geq 1} \frac{2^j}{j} [w^{j-1}] \log(1 + q(2\ell + w))^j.$$

Using (2.12), the coefficient of q^m is

$$\frac{1}{m!} \sum_{j=1}^m 2^j (j-1)! s(m, j) [w^{j-1}] (2\ell + w)^m,$$

which is (7.11). □

7.4 The dominant-core correction

Write

$$\mu_0 = \rho + U(\rho^{-1}), \quad U(s) = \sum_{m \geq 1} a_m s^m. \quad (7.14)$$

Theorem 7.4 (All algebraic inverse coefficients). *The series $U(s)$ is the unique analytic solution near $s = 0$ of*

$$U = 3 \log(1 + sU) - \log(1 - s + sU). \quad (7.15)$$

Its coefficients have the finite Lagrange representation

$$a_m = [s^m] \sum_{q=1}^m \frac{1}{q} [w^{q-1}] (3 \log(1 + sw) - \log(1 - s + sw))^q. \quad (7.16)$$

Equivalently, with $\widehat{B}$ defined by (2.9), they satisfy the triangular recurrence

$$\begin{aligned} a_m = & 3 \sum_{q=1}^{\lfloor m/2 \rfloor} \frac{(-1)^{q+1}}{q} \widehat{B}_{m-q,q}(a_1, a_2, \dots) \\ & + \sum_{q=1}^m \frac{1}{q} \sum_{j=0}^{\min(q, m-q)} (-1)^j \binom{q}{j} \widehat{B}_{m-q,j}(a_1, a_2, \dots). \end{aligned} \quad (7.17)$$

Proof. Set $\mu_0 = \rho + u$ in $F_0(\mu_0) = \rho - 2 \log \rho$. With $s = 1/\rho$,

$$0 = u - 2 \log(1 + su) + \log \left(1 - \frac{s}{1 + su} \right).$$

Separating the last logarithm gives (7.15). The analytic implicit function theorem applies because the derivative with respect to U at $(s, U) = (0, 0)$ is 1.

Introduce a bookkeeping parameter z on the right of (7.15) and apply (2.13); this proves (7.16). Finally expand

$$3 \log(1 + sU) = 3 \sum_{q \geq 1} \frac{(-1)^{q+1}}{q} (sU)^q$$

and

$$-\log(1 - s + sU) = \sum_{q \geq 1} \frac{s^q}{q} (1 - U)^q.$$

Extracting $[s^m]$ and using ordinary Bell polynomials gives (7.17). Every term on the right involves only $a_1, \dots, a_{m-1}$. □

The first coefficients are

$$\begin{aligned} a_1 = 1, \quad a_2 = \frac{5}{2}, \quad a_3 = \frac{13}{3}, \quad a_4 = \frac{53}{12}, \quad a_5 = -\frac{14}{5}, \\ a_6 = -\frac{763}{30}, \quad a_7 = -\frac{2179}{35}, \quad a_8 = -\frac{42289}{630}, \quad a_9 = \frac{132973}{1260}, \quad a_{10} = \frac{3449711}{5040}. \end{aligned} \quad (7.18)$$

In particular,

$$\mu_0 = \rho + \rho^{-1} + \frac{5}{2}\rho^{-2} + \frac{13}{3}\rho^{-3} + \frac{53}{12}\rho^{-4} - \frac{14}{5}\rho^{-5} + \dots \quad (7.19)$$

7.5 The algebraic inverse in the original index

Substituting (7.14) into (7.8) gives

$$\begin{aligned} N_0(y) = \frac{3\rho^2}{2\pi^2} + \frac{1}{24} + \frac{3}{\pi^2} + \frac{15}{2\pi^2\rho} + \frac{29}{2\pi^2\rho^2} + \frac{83}{4\pi^2\rho^3} \\ + \frac{559}{40\pi^2\rho^4} - \frac{611}{20\pi^2\rho^5} - \frac{112459}{840\pi^2\rho^6} + \dots \end{aligned} \quad (7.20)$$

More generally, for $m \geq 1$, the coefficient of ρ^{-m} is

$$\frac{3}{2\pi^2} \left(2a_{m+1} + \sum_{j=1}^{m-1} a_j a_{m-j} \right). \quad (7.21)$$

In elementary logarithms, retaining the first complete block gives

$$\begin{aligned} N_0(y) = \frac{3}{2\pi^2} L^2 + \frac{6}{\pi^2} L \log L + \frac{6}{\pi^2} (\log L)^2 + \frac{12}{\pi^2} \log L \\ + \frac{3}{\pi^2} + \frac{1}{24} + O\left(\frac{(\log L)^2}{L}\right), \quad L = \log(y/\kappa). \end{aligned} \quad (7.22)$$

The exact Lambert form (7.9) plus the convergent U series is numerically preferable to expanding everything in L .

7.6 Phase locking

Let the uncorrected carrier index be

$$N_\rho(y) := \frac{1}{24} + \frac{3\rho(y)^2}{2\pi^2}. \quad (7.23)$$

Then (7.20) implies

$$N_0(y) - N_\rho(y) = \frac{3}{\pi^2} + O(\rho^{-1}). \quad (7.24)$$

The limiting shift is $3/\pi^2 = 0.3039635509\dots$

Warning 7.5 (Do not expand the arithmetic phase about N_ρ). The amplitudes $\mathcal{A}_k(N)$ are periodic on the N scale. Although $\mu_0 - \rho = O(\rho^{-1})$, the induced index change is $N_0 - N_\rho = O(1)$ because $dN/d\mu \asymp \rho$. Taylor expansion of $\mathcal{A}_k(N(\mu_0))$ around ρ therefore does not begin with a small phase perturbation. The full algebraic core N_0 must be used as the base phase. Only the subsequent exponentially small displacement can be expanded.

8 The full inverse transseries

8.1 One-stage universal Lagrange formula

Let $\mathcal{R}(\mu)$ be the exact paired correction (4.6), with A_k replaced by the real amplitudes $\mathcal{A}_k$. Put

$$\mu = \rho + u. \quad (8.1)$$

Taking logarithms in (7.2) and subtracting (7.6) gives

$$0 = u - 2 \log \left(1 + \frac{u}{\rho} \right) + \log \left(1 - \frac{1}{\rho + u} \right) + \log(1 + \mathcal{R}(\rho + u)). \quad (8.2)$$

Define

$$\Phi_\rho(w) := 2 \log \left(1 + \frac{w}{\rho} \right) - \log \left(1 - \frac{1}{\rho + w} \right) - \log(1 + \mathcal{R}(\rho + w)). \quad (8.3)$$

Theorem 8.1 (Complete one-stage inverse formula). *As a formal Rademacher transseries, and analytically for sufficiently large y under the paired-summation convention,*

$$u = \sum_{q \geq 1} \frac{1}{q} [w^{q-1}] \Phi_\rho(w)^q, \quad \mathcal{N}(y) = \frac{1}{24} + \frac{3(\rho + u)^2}{2\pi^2}. \quad (8.4)$$

Every prescribed coefficient of a finite- K , finite-action truncation receives only finitely many contributions.

Proof. Introduce $u = z\Phi_\rho(u)$. Lagrange–Bürmann gives $[z^q]u = q^{-1}[w^{q-1}]\Phi_\rho(w)^q$. Setting $z = 1$ gives the formal identity. For large ρ , the algebraic part is $O(\rho^{-1})$ and $\mathcal{R} = O(\rho^2 e^{-\rho/2})$; on a fixed small disk the map is a contraction, which justifies the analytic branch. $\square$

Formula (8.4) is compact but mixes algebraic and exponential corrections. The two-stage form below exposes every nonperturbative coefficient explicitly and automatically respects phase locking.

8.2 Two-stage expansion about the dominant core

Write

$$\mu = \mu_0 + v. \quad (8.5)$$

Set

$$G(w) := \begin{cases} \frac{F_0(\mu_0 + w) - F_0(\mu_0)}{w}, & w \neq 0, \\ F'_0(\mu_0), & w = 0. \end{cases} \quad (8.6)$$

Then the exact correction satisfies

$$v = -\frac{\log(1 + \mathcal{R}(\mu_0 + v))}{G(v)}. \quad (8.7)$$

Consequently,

$$v = \sum_{q \geq 1} \frac{1}{q} [w^{q-1}] \left(-\frac{\log(1 + \mathcal{R}(\mu_0 + w))}{G(w)} \right)^q. \quad (8.8)$$

This is the phase-locked complete inverse transseries: all arithmetic amplitudes are based at $N_0 = N(\mu_0)$.

8.3 Resolved finite- K variables

For a sector cutoff $K \geq 2$, let

$$\mathcal{J}_K := \{(1, -)\} \cup \{(k, \sigma) : 2 \leq k \leq K, \sigma = \pm 1\}. \quad (8.9)$$

For $j = (k, \sigma) \in \mathcal{J}_K$, set

$$\alpha_j = 1 - \frac{\sigma}{k}, \quad E_j := e^{-\alpha_j \mu_0}, \quad B_j(w) := r_{k, \sigma}(\mu_0 + w) e^{-\alpha_j w}, \quad (8.10)$$

where now

$$r_{k, \sigma}(\mu) = \frac{\mathcal{A}_k(N(\mu))}{\sqrt{k}} \frac{1 - \sigma k / \mu}{1 - 1/\mu}. \quad (8.11)$$

Introduce independent formal variables $\mathbf{E} = (E_j)_{j \in \mathcal{J}_K}$ and

$$S(w; \mathbf{E}) := \sum_{j \in \mathcal{J}_K} E_j B_j(w), \quad Q(w; \mathbf{E}) := \log(1 + S(w; \mathbf{E})). \quad (8.12)$$

For a finite multi-index $\mathbf{m} = (m_j)$, write $|\mathbf{m}| = \sum_j m_j$, $\mathbf{m}! = \prod_j m_j!$, and $\mathbf{E}^{\mathbf{m}} = \prod_j E_j^{m_j}$.

Lemma 8.2 (Coefficients of the logarithm). *For $\mathbf{m} \neq \mathbf{0}$,*

$$[\mathbf{E}^{\mathbf{m}}] Q(w; \mathbf{E}) = (-1)^{|\mathbf{m}|+1} \frac{(|\mathbf{m}| - 1)!}{\mathbf{m}!} \prod_{j \in \mathcal{J}_K} B_j(w)^{m_j}. \quad (8.13)$$

Proof. Only the term $S^{|\mathbf{m}|}$ in $\log(1 + S) = \sum_{d \geq 1} (-1)^{d+1} S^d / d$ contributes. Its multinomial coefficient is $|\mathbf{m}|! / \mathbf{m}!$. $\square$

8.4 General multi-sector coefficient formula

Expand

$$v = \sum_{\mathbf{m} \neq \mathbf{0}} V_{\mathbf{m}}(\mu_0) \mathbf{E}^{\mathbf{m}}. \quad (8.14)$$

Theorem 8.3 (All inverse-sector coefficients). *For every nonzero multi-index $\mathbf{m}$,*

$$V_{\mathbf{m}} = [\mathbf{E}^{\mathbf{m}}] \sum_{q=1}^{|\mathbf{m}|} \frac{1}{q} [w^{q-1}] \left(-\frac{Q(w; \mathbf{E})}{G(w)} \right)^q. \quad (8.15)$$

Equivalently, if $q_{\mathbf{a}}(w) := [\mathbf{E}^{\mathbf{a}}] Q(w; \mathbf{E})$ is given by (8.13), then

$$V_{\mathbf{m}} = \sum_{q=1}^{|\mathbf{m}|} \frac{(-1)^q}{q} [w^{q-1}] \frac{1}{G(w)^q} \sum_{\substack{\mathbf{m}_1 + \dots + \mathbf{m}_q = \mathbf{m} \\ \mathbf{m}_i \neq \mathbf{0}}} \prod_{i=1}^q q_{\mathbf{m}_i}(w). \quad (8.16)$$

Both sums are finite.

Proof. Apply Lagrange–Bürmann to $v = z[-Q(v; \mathbf{E})/G(v)]$ and extract $\mathbf{E}^{\mathbf{m}}$. Since every monomial of Q has positive total $\mathbf{E}$ -degree, only $q \leq |\mathbf{m}|$ contributes. Expanding Q^q gives (8.16). $\square$

This is the promised general coefficient formula for the full inverse transseries. It consists solely of finite coefficient extractions, multinomial products, and derivatives at $w = 0$; those derivatives may in turn be written with complete Bell polynomials.

8.5 Triangular recursion and low product orders

Let $D_0 := F'_0(\mu_0)$. Substitution of (8.14) into

$$D_0 v + \sum_{r \geq 2} \frac{F_0^{(r)}(\mu_0)}{r!} v^r + Q(v; \mathbf{E}) = 0 \quad (8.17)$$

provides a triangular alternative to Theorem 8.3:

$$V_{\mathbf{m}} = -\frac{1}{D_0} [E^{\mathbf{m}}] \left[\sum_{r \geq 2} \frac{F_0^{(r)}(\mu_0)}{r!} v^r + Q(v; \mathbf{E}) \right]_{\text{lower total degree}}. \quad (8.18)$$

The bracket at total degree $|\mathbf{m}|$ contains no unknown $V_{\mathbf{m}}$.

For a unit multi-index $\mathbf{e}_j$,

$$V_{\mathbf{e}_j} = -\frac{r_j(\mu_0)}{F'_0(\mu_0)}. \quad (8.19)$$

Writing $B_j = B_j(0)$ and $B'_j = B'_j(0)$, the repeated-sector coefficient is

$$V_{2\mathbf{e}_j} = -\frac{1}{D_0} \left[\frac{1}{2} F''_0(\mu_0) V_{\mathbf{e}_j}^2 + B'_j V_{\mathbf{e}_j} - \frac{1}{2} B_j^2 \right]. \quad (8.20)$$

For distinct j and ℓ ,

$$V_{\mathbf{e}_j + \mathbf{e}_\ell} = -\frac{1}{D_0} \left[F''_0(\mu_0) V_{\mathbf{e}_j} V_{\mathbf{e}_\ell} + B'_j V_{\mathbf{e}_\ell} + B'_\ell V_{\mathbf{e}_j} - B_j B_\ell \right]. \quad (8.21)$$

These formulae display explicitly how nonlinear reversion mixes arithmetic sectors.

8.6 Coefficients for the inverse index

Since

$$\mathcal{N}(y) = N_0(y) + \frac{3\mu_0}{\pi^2} v + \frac{3}{2\pi^2} v^2, \quad (8.22)$$

the coefficient of $E^{\mathbf{m}}$ in the inverse index is

$$\mathfrak{N}_{\mathbf{m}} = \frac{3\mu_0}{\pi^2} V_{\mathbf{m}} + \frac{3}{2\pi^2} \sum_{\substack{\mathbf{a} + \mathbf{b} = \mathbf{m} \\ \mathbf{a}, \mathbf{b} \neq \mathbf{0}}} V_{\mathbf{a}} V_{\mathbf{b}}. \quad (8.23)$$

Equations (8.15) and (8.23) are an all-order, closed coefficient calculus for the inverse.

8.7 Primary inverse sectors

At first nonperturbative order, (8.19) and (8.22) give

$$\Delta N_{k, \sigma}^{(1)}(y) = -\frac{3\mu_0}{\pi^2 F'_0(\mu_0)} \frac{\mathcal{A}_k(N_0)}{\sqrt{k}} \frac{1 - \sigma k / \mu_0}{1 - 1/\mu_0} e^{-(1 - \sigma/k)\mu_0}. \quad (8.24)$$

The first two positive sectors are

$$\Delta N_{2, +}^{(1)} = -\frac{3\mu_0}{\pi^2 F'_0(\mu_0)} \frac{\cos(\pi N_0)}{\sqrt{2}} \frac{1 - 2/\mu_0}{1 - 1/\mu_0} e^{-\mu_0/2}, \quad (8.25)$$

$$\Delta N_{3,+}^{(1)} = -\frac{3\mu_0}{\pi^2 F'_0(\mu_0)} \frac{2}{\sqrt{3}} \cos(\pi N_0) \cos\left(\frac{\pi}{18} + \frac{\pi N_0}{3}\right) \frac{1-3/\mu_0}{1-1/\mu_0} e^{-2\mu_0/3}. \quad (8.26)$$

Thus, before the accumulation action 1,

$$\mathcal{N}(y) = N_0(y) + \Delta N_{2,+}^{(1)}(y) + \Delta N_{3,+}^{(1)}(y) + \mathcal{O}\left(\mu_0^3 e^{-3\mu_0/4}\right). \quad (8.27)$$

The displayed remainder is a convenient coarse bound obtained from [Theorem 4.2](#); individual resolved terms are generally smaller by powers of μ_0 .

8.8 Fractional powers of the inverse argument

The dominant-core equation gives the exact conversion

$$e^{-\alpha\mu_0} = \left(\frac{\kappa}{y}\right)^\alpha \mu_0^{-2\alpha} \left(1 - \frac{1}{\mu_0}\right)^\alpha. \quad (8.28)$$

Hence a primary positive k -sector has the scale

$$\Delta N_{k,+}^{(1)} = y^{-(1-1/k)} \mu_0^{-1+2/k} \times \{\text{bounded arithmetic amplitude and a } \mu_0^{-1} \text{ series}\}. \quad (8.29)$$

The staircase begins with

$$y^{-1/2}, \quad y^{-2/3}(\log y)^{-1/3}, \quad y^{-3/4}(\log y)^{-1/2}, \dots,$$

and accumulates at the y^{-1} level.

For $k = 2$, (8.25) becomes

$$\Delta N_{2,+}^{(1)} = -\frac{3\sqrt{k}}{\pi^2 \sqrt{2}} \frac{\cos(\pi N_0)}{\sqrt{y}} \frac{1-2/\mu_0}{F'_0(\mu_0) \sqrt{1-1/\mu_0}}. \quad (8.30)$$

Therefore

$$\boxed{\mathcal{N}(y) = N_0(y) - \frac{3\sqrt{k}}{\pi^2 \sqrt{2}} \frac{\cos(\pi N_0(y))}{\sqrt{y}} \left(1 + \mathcal{O}\left(\frac{1}{\log y}\right)\right) + \tilde{\mathcal{O}}(y^{-2/3})}. \quad (8.31)$$

Here $\tilde{\mathcal{O}}$ permits powers of $\log y$. The constant is

$$\frac{3\sqrt{k}}{\pi^2 \sqrt{2}} = 0.2094596846361762626 \dots \quad (8.32)$$

Because $N_0(y) \asymp (\log y)^2$, the cosine in (8.31) is a quadratic logarithmic chirp, not a log-periodic oscillation.

8.9 Completion at the accumulation action

For any K , [Theorem 8.3](#) resolves all sectors in $\mathcal{J}_K$ and all of their products. The omitted primary tail is kept in paired form:

$$\mathcal{R}_{>K}(\mu) = \frac{2e^{-\mu}}{1-1/\mu} \sum_{k>K} \frac{\mathcal{A}_k(N(\mu))}{\sqrt{k}} g\left(\frac{\mu}{k}\right). \quad (8.33)$$

One may insert (8.33) as an additional analytic symbol in (8.8). Increasing K produces a compatible transasymptotic resolution, while (4.9) controls the unresolved part. This is the precise meaning of the full inverse at and beyond action 1.

9 Residual transfer and rigorous truncation control

Formal coefficient matching does not by itself bound the error of an actual inverse. The needed analytic step is elementary but should be stated.

Theorem 9.1 (Residual-to-inverse error transfer). *Let*

$$H(\mu; y) := F_0(\mu) + \log(1 + \mathcal{R}(\mu)) - \log(y/\kappa). \quad (9.1)$$

Suppose $H(\mu; y) = 0$ has a solution μ_ in an interval I , and*

$$\inf_{\mu \in I} |\partial_\mu H(\mu; y)| \geq m > 0. \quad (9.2)$$

If an approximation $\tilde{\mu} \in I$ has residual $\varepsilon = H(\tilde{\mu}; y)$ and the line segment joining $\tilde{\mu}$ to μ_ lies in I , then*

$$|\tilde{\mu} - \mu_*| \leq \frac{|\varepsilon|}{m}. \quad (9.3)$$

Consequently,

$$|N(\tilde{\mu}) - N(\mu_*)| \leq \frac{3}{2\pi^2} (2|\mu_*| + |\tilde{\mu} - \mu_*|) \frac{|\varepsilon|}{m}. \quad (9.4)$$

Proof. Apply the mean-value theorem to H and then to $N(\mu) = 1/24 + 3\mu^2/(2\pi^2)$. $\square$

For large μ , $F'_0(\mu) = 1 + O(\mu^{-1})$ and $\mathcal{R}'(\mu)$ is exponentially small up to a polynomial factor, so one may take, for example, $m = 1/2$ beyond an effectively checkable threshold. This turns any formal truncation into a certificate once its residual is bounded.

9.1 Algebraic truncation

If

$$U_M(\rho^{-1}) = \sum_{m=1}^M a_m \rho^{-m}, \quad (9.5)$$

then the implicit analytic equation (7.15) gives

$$\mu_0 - (\rho + U_M) = O(\rho^{-M-1}), \quad (9.6)$$

and therefore

$$N_0 - \left[\frac{1}{24} + \frac{3}{2\pi^2} (\rho + U_M)^2 \right] = O(\rho^{-M}). \quad (9.7)$$

To evaluate an arithmetic amplitude to $o(1)$ phase error, at least the constant index shift in (7.24) must be included.

9.2 Sector truncation

For a fixed K , substitute the resolved finite sum and paired tail bound into (9.1). Since $\log(1+z) = z + O(z^2)$ for small z , the direct bound (4.9) transfers to the inverse with one additional factor of order μ in the original index. In particular, for fixed K ,

$$\mathcal{N}(y) - \mathcal{N}_K(y) = O_K \left(\mu_0^3 \exp \left[- \left(1 - \frac{1}{K+1} \right) \mu_0 \right] \right), \quad (9.8)$$

where $\mathcal{N}_K$ is the inverse obtained from sectors through $k = K$ and the $k = 1$ negative sector. Sharper bounds follow by using nontrivial estimates for A_k or by isolating the first omitted k .

9.3 Certified recovery of an integer index

For $y = p(n)$, any approximation $\tilde{N}(y)$ satisfying

$$|\tilde{N}(y) - n| < \frac{1}{2} \quad (9.9)$$

recovers n by nearest-integer rounding. The dominant-core inverse $N_0(y)$ already has exponentially small error, so (9.9) holds for all sufficiently large n . Adding the first few primary sectors makes the margin enormous even at moderate n .

10 Algorithms

All formulae in this article are constructive. This section records concise implementations independent of any one computer algebra system.

10.1 Direct sector coefficients

Algorithm 10.1 (Finite-sum generation of $C_{k,\sigma;r}$). Given $k \geq 1$, $\sigma \in \{\pm 1\}$, and R :

1. Set $b = \sigma\pi/(6k)$.
2. For $0 \leq r \leq R$, compute

$$C_r = (-4\sqrt{6})^{-r} \sum_{j=0}^{\lfloor (r+1)/2 \rfloor} \binom{r+1}{j} \frac{r+1-j}{(r+1-2j)!} b^{r-2j}.$$

3. Use (5.2), pairing the two signs before summing over large k .

The arithmetic complexity for one coefficient is $O(r)$ operations in the coefficient field.

10.2 Lambert carrier polynomials

Algorithm 10.2 (Generation of $P_m(\ell)$). Precompute signed Stirling numbers $s(m, j)$ by

$$s(m+1, j) = s(m, j-1) - m s(m, j), \quad s(0, 0) = 1.$$

Then apply (7.11). This gives the full elementary-logarithmic expansion (7.10); for numerical work, evaluate the exact W_{-1} expression instead.

10.3 Dominant-core coefficients

Algorithm 10.3 (Triangular generation of a_m). Initialize $a_1 = 1$. For $m = 2, 3, \dots, M$, compute the ordinary Bell coefficients $\hat{B}_{n,j}$ from the current truncated series $U(s) = \sum_{r < m} a_r s^r$, and use (7.17). No nonlinear solve is required at any stage.

A direct symbolic alternative is the following pseudo-code.

```

U = 0
for m = 1..M:
    Phi(w,s) = 3*log(1+s*w) - log(1-s+s*w)
    a[m] = coeff(s,m, sum(q=1..m, coeff(w,q-1, Phi(w,s)^q)/q))
    U = U + a[m]*s^m

```

10.4 Full multi-sector inverse

Algorithm 10.4 (Coefficient of a prescribed multi-index). Fix K and a nonzero multi-index $\mathbf{m}$.

1. Compute ρ from (7.9), then μ_0 from the convergent U series or one Newton step on (7.7).
2. Form $G(w)$, $B_j(w)$, and $q_a(w)$ using (8.6), (8.10), and (8.13).
3. Evaluate the finite sum (8.16); only $1 \leq q \leq |\mathbf{m}|$ and Taylor order $q - 1$ are needed.
4. Convert $V_{\mathbf{m}}$ to the inverse-index coefficient with (8.23).

For many coefficients, the triangular recurrence (8.18) avoids repeated composition and is preferable.

10.5 Numerical inversion

For a numerical value y :

1. compute ρ by W_{-1} ;
2. compute μ_0 from (7.7);
3. evaluate a finite paired Rademacher correction $\mathcal{R}_K$;
4. solve $F_0(\mu) + \log(1 + \mathcal{R}_K(\mu)) = \log(y/\kappa)$ by Newton's method;
5. return $N(\mu)$, and use (6.6) for the sequence inverse.

The derivative is asymptotic to 1 in the μ coordinate, so the iteration is well conditioned.

11 Numerical validation

All computations in this section used integer partition values generated by Euler's pentagonal recurrence, exact rational Dedekind sums, and 80-decimal floating-point evaluation of the transcendental factors. The tables are intended as independent checks of normalization, sector ordering, and inverse signs; they are not used in any proof.

11.1 Forward expansion

Let $H(n)$ be the elementary Hardy–Ramanujan term (1.2), let $D(n) = P_{1,+}(n)$, and let $S_K(n)$ be the paired Rademacher sum through $k = K$. The entries below are relative errors $X(n)/p(n) - 1$.

Table 1: Relative errors of direct approximations.

n	H	D	S_2	S_3	S_5
10	1.4534069×10^{-1}	$-8.8622418 \times 10^{-3}$	1.6605794×10^{-3}	3.7391400×10^{-4}	3.4721733×10^{-4}
50	6.5439754×10^{-2}	$-7.3073551 \times 10^{-5}$	3.9027635×10^{-6}	2.1049640×10^{-7}	$-1.1350427 \times 10^{-7}$
100	4.5713563×10^{-2}	$-1.8219969 \times 10^{-6}$	8.6837718×10^{-9}	$-4.9491206 \times 10^{-9}$	$3.1453999 \times 10^{-10}$
500	2.0094903×10^{-2}	$-2.4389452 \times 10^{-13}$	$1.7536653 \times 10^{-17}$	$-2.0016640 \times 10^{-19}$	$-3.5933306 \times 10^{-22}$
1000	1.4152439×10^{-2}	$-1.6997051 \times 10^{-18}$	$1.2573854 \times 10^{-24}$	$-3.4743827 \times 10^{-27}$	$3.9452980 \times 10^{-30}$

The contrast between H and D confirms that the full algebraic half-power series belongs to the single shifted dominant sector. The remaining error is exponentially small and alternates with the arithmetic amplitudes.

11.2 Inverse expansion and phase locking

For $y = p(n)$, define four approximations:

- (i) $N_\rho = 1/24 + 3\rho^2/(2\pi^2)$;
- (ii) N_0 , the exact dominant-core inverse;
- (iii) $N_{\text{lin},10} := N_0 + (3\mu_0/\pi^2)v_{\text{lin}}$, where $v_{\text{lin}} = -\mathcal{R}_{10}(\mu_0)/F'_0(\mu_0)$; this retains only the first-order inverse-index coefficient;
- (iv) N_{20} , the Newton inverse of the paired interpolation truncated at $k = 20$.

The table lists approximation minus n .

Table 2: Errors in the inverse at exact partition values $y = p(n)$.

n	$N_p - n$	$N_0 - n$	$N_{\text{lin},10} - n$	$N_{20} - n$
10	$-3.9910002 \times 10^{-1}$	2.8447205×10^{-2}	$-8.0725131 \times 10^{-4}$	$-1.0235155 \times 10^{-4}$
50	$-3.5044252 \times 10^{-1}$	4.5102431×10^{-4}	$-9.9453881 \times 10^{-8}$	1.9795735×10^{-8}
100	$-3.3599875 \times 10^{-1}$	1.5378046×10^{-5}	$-5.2359563 \times 10^{-11}$	$7.2911158 \times 10^{-11}$
500	$-3.1767515 \times 10^{-1}$	$4.4042295 \times 10^{-12}$	$1.1396547 \times 10^{-23}$	$1.6512140 \times 10^{-23}$
1000	$-3.1356078 \times 10^{-1}$	$4.2959998 \times 10^{-17}$	$-3.8971591 \times 10^{-31}$	3.37×10^{-35}

The first column tends slowly toward $-3/\pi^2$, exactly as predicted by (7.24). The dominant-core correction removes that $O(1)$ phase error, after which the residual is of Rademacher size. Linearized nonperturbative inversion already captures many additional digits.

11.3 Parity check

At $n = 49, 50, 51$, the dominant-core errors are respectively

$$-5.2643134 \times 10^{-4}, \quad 4.5102431 \times 10^{-4}, \quad -4.0920475 \times 10^{-4}.$$

Their alternating sign is explained by $\mathcal{A}_2(n) = (-1)^n$. After applying only the $k = 2, +$ correction (8.25), the errors become

$$-1.0889617 \times 10^{-5}, \quad -2.4087145 \times 10^{-5}, \quad 2.9028845 \times 10^{-5},$$

and adding the remaining linear sectors through $k = 10$ reduces them to -2.24×10^{-7} , -9.95×10^{-8} , and -5.81×10^{-8} . This verifies both the sign and the phase of the leading inverse chirp.

12 Interpretation and further deductions

12.1 A two-level transseries geometry

The direct expansion has a nested geometry. At the outer level, the dominant sector is multiplied by $1 + \mathcal{R}$. At the inner level, the coefficient of the condensed factor $e^{-\mu}$ in (4.6) contains $e^{\pm\mu/k}$ and is resolved by increasing k . This is analogous to a transasymptotic expansion in which an infinite family of saddles condenses at a limiting action. The pairing constraint is part of that geometry.

The inverse inherits the same structure. Product sectors have actions

$$\alpha(\mathbf{m}) = \sum_{j \in \mathcal{J}_K} m_j \alpha_j. \quad (12.1)$$

Below action 1, only the primary positive sectors occur: any product of two positive actions is at least 1. At action 1, nonlinear products such as $(k = 2, +)^2$ appear at the same limiting level approached by the infinitely many primary positive sectors. This is where the paired completion becomes essential.

12.2 Arithmetic phase as a coefficient algebra

After phase locking, the coefficient algebra is generated by rational functions of μ_0 and the finite trigonometric polynomials

$$\mathcal{A}_k(N_0) = \sum_{(h,k)=1} \cos\left(\pi s(h, k) - \frac{2\pi h}{k} \left(\frac{1}{24} + \frac{3\mu_0^2}{2\pi^2}\right)\right). \quad (12.2)$$

Differentiation with respect to μ_0 preserves this algebra after adjoining polynomial factors in μ_0 . Therefore every Taylor coefficient in Theorem 8.3 remains explicit. In terms of y , the phases are quadratic in $\log y$ to leading order.

12.3 Why the inverse is unusually accurate

The leading partition value grows like $e^{c\sqrt{n}}$, so a relative error ε in $p(n)$ induces an index error of order $\sqrt{n}\varepsilon$. The first omitted direct sector after the dominant core has relative size $e^{-c\sqrt{n}/2}$. Hence the dominant-core inverse at $y = p(n)$ differs from n by approximately

$$O\left(\sqrt{n}e^{-c\sqrt{n}/2}\right),$$

which rapidly becomes much smaller than one. This explains why rounding a carefully shifted inverse can recover the exact index far earlier than the slow relative convergence of the elementary Hardy–Ramanujan approximation might suggest.

12.4 Generalization template

The same method applies to a broad class of coefficient sequences with a Rademacher- or saddle-sector representation

$$a(n) = Ke^{\Phi(n)}\Phi(n)^{-\gamma}\left(A_0(\Phi(n)) + \sum_j A_j(n, \Phi(n))e^{-\alpha_j\Phi(n)}\right). \quad (12.3)$$

The reusable steps are:

1. choose a phase variable that makes each saddle prefactor elementary;
2. derive all algebraic coefficients with Bell polynomials;
3. invert the dominant exponential-power carrier with Lambert W or its generalized analogue;
4. resum every $O(1)$ displacement in any periodic coefficient phase;
5. apply the multi-index Lagrange formula to the exponentially small sectors;
6. retain a paired or integral completion whenever saddle actions accumulate.

13 Conclusion

The partition numbers possess a particularly transparent complete asymptotic structure once the modular shift $n - 1/24$ is used. Every fixed Rademacher sector is exactly a two-term exponential expression, and its entire $n^{-1/2}$ coefficient sequence is available in the finite formula (5.5). The ordinary Hardy–Ramanujan Poincaré series is only the algebraic expansion of the dominant positive sector; the rest of Rademacher’s convergent series is its nonperturbative completion.

For inversion, the exact Lambert carrier (7.9), the all-order algebraic correction (7.16)–(7.17), and the multi-index nonperturbative formula (8.15) provide a complete coefficient calculus. The arithmetic amplitudes force a phase-locked organization based at $N_0(y)$ rather than the raw Lambert index. The first correction is the parity chirp (8.31); subsequent primary sectors form a fractional-power staircase accumulating at y^{-1} . The paired Rademacher tail completes the expansion at that accumulation level.

The resulting framework is both symbolic and analytic: coefficients are finite combinatorial expressions, while residual transfer and the explicit paired-tail estimate convert truncations into controlled approximations. It also gives a precise bridge from a smooth inverse to the generalized inverse of the discrete sequence through the exact ceiling relation (6.6).

A Residue proof of the finite sector coefficient formula

This appendix supplies the promised detailed proof of Theorem 5.2. Start from the generating function (5.3). Put

$$u = -\frac{t}{4\sqrt{6}}, \quad b = \frac{\sigma\pi}{6k}. \quad (A.1)$$

Because $\sigma c/k = 2\sqrt{6}b$, one obtains

$$G(-4\sqrt{6}u) := \sum_{r \geq 0} C_{k,\sigma;r} (-4\sqrt{6}u)^r = \frac{e^{bT(u)}}{1-4u^2} \left(1 + \frac{2u}{b\sqrt{1-4u^2}} \right), \quad (\text{A.2})$$

where

$$T(u) := \frac{1 - \sqrt{1-4u^2}}{2u} \quad (\text{A.3})$$

satisfies

$$T = u(1+T^2), \quad u = \frac{T}{1+T^2}, \quad \sqrt{1-4u^2} = \frac{1-T^2}{1+T^2}. \quad (\text{A.4})$$

Differentiating the last expression for u gives

$$du = \frac{1-T^2}{(1+T^2)^2} dT. \quad (\text{A.5})$$

Substitution into (A.2) yields the exact differential identity

$$\begin{aligned} G(-4\sqrt{6}u) du &= e^{bT} \left(\frac{1}{1-T^2} + \frac{2T}{b(1-T^2)^2} \right) dT \\ &= \frac{1}{b} d \left(\frac{e^{bT}}{1-T^2} \right). \end{aligned} \quad (\text{A.6})$$

Let

$$D_r := C_{k,\sigma;r} (-4\sqrt{6})^r = [u^r] G(-4\sqrt{6}u). \quad (\text{A.7})$$

Residue transfer gives

$$\begin{aligned} D_r &= \text{Res}_{u=0} G(-4\sqrt{6}u) u^{-r-1} du \\ &= \frac{1}{b} \text{Res}_{T=0} T^{-r-1} (1+T^2)^{r+1} d \left(\frac{e^{bT}}{1-T^2} \right). \end{aligned} \quad (\text{A.8})$$

The residue of a derivative is zero, so integration by parts gives

$$\begin{aligned} D_r &= -\frac{1}{b} \text{Res}_{T=0} \frac{e^{bT}}{1-T^2} d \left(T^{-r-1} (1+T^2)^{r+1} \right) \\ &= \frac{r+1}{b} \text{Res}_{T=0} e^{bT} (1+T^2)^r T^{-r-2} dT \\ &= \frac{r+1}{b} [T^{r+1}] e^{bT} (1+T^2)^r. \end{aligned} \quad (\text{A.9})$$

Expanding the exponential and binomial gives

$$D_r = \frac{r+1}{b} \sum_{j=0}^{\lfloor (r+1)/2 \rfloor} \binom{r}{j} \frac{b^{r+1-2j}}{(r+1-2j)!}. \quad (\text{A.10})$$

Finally,

$$(r+1) \binom{r}{j} = (r+1-j) \binom{r+1}{j}, \quad (\text{A.11})$$

and division by $(-4\sqrt{6})^r$ proves (5.5).

B Additional algebraic inverse coefficients

For reference, the next coefficients after those displayed in (7.18) are

$$a_{11} = \frac{29813143}{18480}, \quad a_{12} = \frac{496569589}{356400}. \quad (\text{B.1})$$

The corresponding continuation of (7.20) is

$$-\frac{101329}{420\pi^2\rho^7} - \frac{228677}{3360\pi^2\rho^8} + O(\rho^{-9}). \quad (\text{B.2})$$

The recurrence (7.17) can be checked directly from the first few ordinary Bell polynomials:

$$\begin{aligned} \widehat{B}_{n,1} &= a_n, \\ \widehat{B}_{n,2} &= \sum_{j=1}^{n-1} a_j a_{n-j}, \\ \widehat{B}_{n,3} &= \sum_{i+j+k=n} a_i a_j a_k, \end{aligned}$$

with all indices positive. This representation is useful for exact rational computation.

C Formula sheet

For convenient reuse, the central formulas are collected here.

Object	Formula
Shift and phase	$x = n - 1/24, \mu = c\sqrt{x}, c = \pi\sqrt{2/3}$
Exact sector	$P_{k,\sigma} = A_k(4\sqrt{3k}x)^{-1}e^{\sigma\mu/k}(1 - \sigma k/\mu)$
Action	$\alpha_{k,\sigma} = 1 - \sigma/k$
Paired kernel	$g(z) = \cosh z - \sinh z/z$
Dominant factor	$D(\mu) = \kappa e^\mu \mu^{-2}(1 - \mu^{-1}), \kappa = \pi^2/(6\sqrt{3})$
Direct coefficient	Equation (5.5)
Bell coefficient	Equation (5.9)
Lambert carrier	$\rho = -2W_{-1}(-\frac{1}{2}\sqrt{\kappa/y})$
Carrier polynomial	Equation (7.11)
Algebraic core	$\mu_0 = \rho + U(\rho^{-1}), \text{ with (7.15)}$
Core coefficient	Equations (7.16) and (7.17)
Full one-stage inverse	Equation (8.4)
Multi-sector coefficient	Equation (8.15)
Inverse-index coefficient	Equation (8.23)
Leading chirp	Equation (8.31)
Discrete inverse	$p_{\mathbb{Z}}^{\langle -1 \rangle}(y) = \lceil \mathcal{N}(y) \rceil$

D Notes on reproducibility

The symbolic identities can be verified with any system that supports formal power series.

1. Expand (5.3) and compare with (5.5); equality through order r uses exact arithmetic in b .
2. Substitute $U = \sum_{m=1}^M a_m s^m$ into (7.15); the residual begins at s^{M+1} .
3. Substitute (7.10) with (7.11) into $\rho - 2 \log \rho - L$; the residual begins at the requested power of L^{-1} .
4. For a finite sector set, substitute coefficients from (8.15) into (8.17); every prescribed multi-index cancels identically.

Numerical partition values may be generated without external tables by Euler's pentagonal recurrence

$$p(n) = \sum_{m \geq 1} (-1)^{m-1} \left[p\left(n - \frac{m(3m-1)}{2}\right) + p\left(n - \frac{m(3m+1)}{2}\right) \right], \quad (\text{D.1})$$

where $p(0) = 1$ and $p(r) = 0$ for $r < 0$.

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